I- Introduction

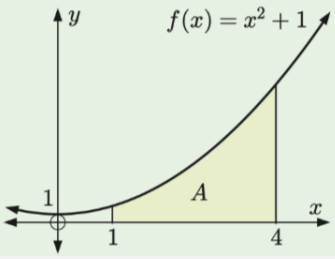
Investigation

The function $f(x) = x^2 + 1$ lies above the x-axis for all $x \in \mathbb{R}$.

Things to think about:

a How can we calculate the shaded area A, which is the area under the curve for $1 \leqslant x \leqslant 4$?

b What function has $x^2 + 1$ as its derivative?

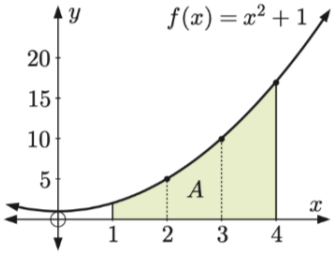


a) Ancient mathematicians way

Consider the function $f(x) = x^2 + 1$.

We wish to estimate the area A enclosed by $y = f(x)$, the x-axis, and the vertical lines $x = 1$ and $x = 4$.

Suppose we divide the interval $1 \leqslant x \leqslant 4$ into three strips of width 1 unit as shown. We obtain three sub-intervals of equal width.



Upper rectangles :

The area of the upper rectangles,

$$A_U = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots \text{ units}^2$$

Lower rectangles :

The area of the lower rectangles,

$$A_L = \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$$
$$= \ldots\ldots \text{ units}^2$$

Now clearly $A_L < A < A_U$, so the area A lies between ……units^2 and ……units^2.

If the interval $1 \leqslant x \leqslant 4$ was divided into 6 subintervals, each of width $\frac{1}{2}$, then

$$\begin{aligned} A_U &= \tfrac{1}{2}f\left(1\tfrac{1}{2}\right) + \tfrac{1}{2}f(2) + \tfrac{1}{2}f\left(2\tfrac{1}{2}\right) + \tfrac{1}{2}f(3) + \tfrac{1}{2}f\left(3\tfrac{1}{2}\right) + \tfrac{1}{2}f(4) \\ &= \tfrac{1}{2}\left(\tfrac{13}{4} + 5 + \tfrac{29}{4} + 10 + \tfrac{53}{4} + 17\right) \\ &= 27.875 \text{ units}^2 \end{aligned}$$

and

$$\begin{aligned} A_L &= \tfrac{1}{2}f(1) + \tfrac{1}{2}f\left(1\tfrac{1}{2}\right) + \tfrac{1}{2}f(2) + \tfrac{1}{2}f\left(2\tfrac{1}{2}\right) + \tfrac{1}{2}f(3) + \tfrac{1}{2}f\left(3\tfrac{1}{2}\right) \\ &= \tfrac{1}{2}\left(2 + \tfrac{13}{4} + 5 + \tfrac{29}{4} + 10 + \tfrac{53}{4}\right) \\ &= 20.375 \text{ units}^2 \end{aligned}$$

From this refinement we conclude that the area A lies between 20.375 and 27.875 units^2.

As we create more subdivisions, the estimates A_L and A_U will become more and more accurate. In fact, as the subdivision width is reduced further and further, both A_L and A_U will converge to A.

Note. *We can use technology to help calculate A_L and A_U for large values of n.*

The example of $y = x^2$ and the x-axis for $0 \leqslant x \leqslant 1$ is of historical interest. Archimedes (287 - 212 BC) found the exact area. In an article that contains 24 propositions, he developed the essential theory of what is now known as integral calculus.

EXERCISE 1

Consider the area between $y = x$ and the x-axis from $x = 0$ to $x = 1$.

a Divide the interval into 5 strips of equal width, then estimate the area using:

i upper rectangles ii lower rectangles.

b Calculate the actual area and compare it with your answers in a.

EXERCISE 2

Consider the area between $y = \dfrac{1}{x}$ and the x-axis from $x = 2$ to $x = 4$. Divide the interval into 6 strips of equal width, then estimate the area using:

a upper rectangles b lower rectangles.

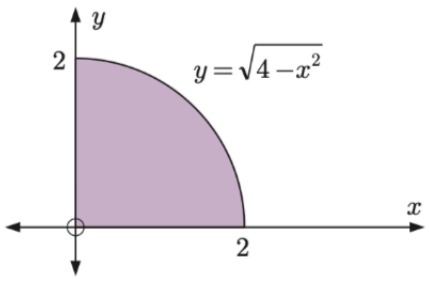
EXERCISE 3

Consider the quarter circle of centre $(0, 0)$ and radius 2 units illustrated.

Its area is $\frac{1}{4}$(full circle of radius 2)

$= \frac{1}{4} \times \pi \times 2^2$

$= \pi$ units^2



a Estimate the area using lower and upper rectangles for $n = 10, 50, 100, 200, 1000$, and $10\,000$. Hence, find rational bounds for π.

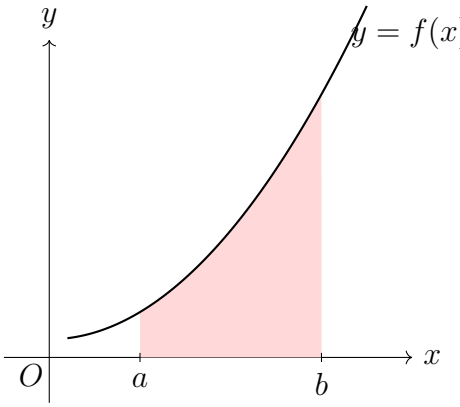
b Archimedes found the famous approximation $3\frac{10}{71} < \pi < 3\frac{1}{7}$.
For what value of n is your estimate for π better than that of Archimedes?

b) Some definitions

Definition

The value A is known as the "definite integral of $f(x)$ from a to b", written $A = \int_a^b f(x)\,dx$.

If $f(x) \geqslant 0$ for all $a \leqslant x \leqslant b$, then $\int_a^b f(x)\,dx$ is equal to the shaded area.



Example

1. Sketch the graph of $y = x^4$ for $0 \leqslant x \leqslant 1$. Shade the area described by $\int_0^1 x^4\,dx$.
2. Use technology to calculate the lower and upper rectangle sums for n equal subdivisions where $n = 5, 10, 50, 100$, and 500.
3. Hence find $\int_0^1 x^4\,dx$ to 2 significant figures.

EXERCISE 4

a Sketch the graph of $y = \sqrt{x}$ for $0 \leqslant x \leqslant 1$.

Shade the area described by $\int_0^1 \sqrt{x}\, dx$.

b Find the lower and upper rectangle sums for $n = 5$, 10, 50, 100, and 500.

c Use the information in b to find $\int_0^1 \sqrt{x}\, dx$ to 2 significant figures.

Example

Use graphical evidence and known area facts to find :

1. $\int_0^2 (2x+1)\, dx$
2. $\int_0^1 \sqrt{1-x^2}\, dx$

EXERCISE 5

Use graphical evidence and known area facts to find:

a $\int_1^3 (1+4x)\, dx$ b $\int_{-1}^2 (2-x)\, dx$ c $\int_{-2}^2 \sqrt{4-x^2}\, dx$

II- Antidifferentiation

Examples of problems we need to solve include :

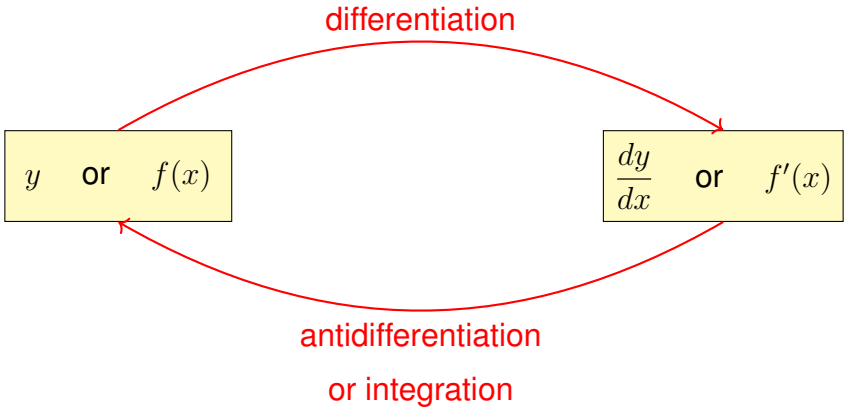
— The gradient function $f'(x)$ of a curve is $2x+3$ and the curve passes through the origin. What is the function $y = f(x)$?

— The rate of change in temperature is $\frac{dT}{dt} = 10e^{-t}$ °C per minute where $t \geqslant 0$. What is the temperature function given that initially the temperature was 11°C ?

a) Definition

Definition

The process of finding y from $\frac{dy}{dx}$, or $f(x)$ from $f'(x)$, is the reverse process of differentiation. We call it antidifferentiation.



Definition

If $F(x)$ is a function where $F'(x) = f(x)$ we say that :

— the derivative of $F(x)$ is $f(x)$ and

— the antiderivative of $f(x)$ is $F(x)$.

Example

Find the antiderivative of :

1. x^3
2. e^{2x}
3. $\frac{1}{\sqrt{x}}$

EXERCISE 6

a Find the antiderivative of:

i x **ii** x^2 **iii** x^5 **iv** x^{-2} **v** x^{-4} **vi** $x^{\frac{1}{3}}$ **vii** $x^{-\frac{1}{2}}$

b From your answers in **a**, predict a general rule for the antiderivative of x^n, for $n \neq -1$.

EXERCISE 7

a Find the antiderivative of:

i e^{2x} **ii** e^{5x} **iii** $e^{\frac{1}{2}x}$ **iv** $e^{0.01x}$ **v** $e^{\pi x}$ **vi** $e^{\frac{x}{3}}$

b From your answers in **a**, predict a general rule for the antiderivative of e^{kx} where k is a constant.

EXERCISE 8

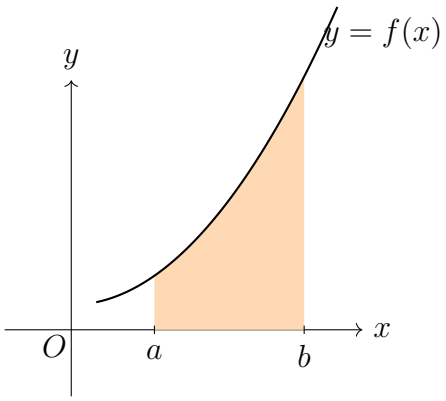
Find the antiderivative of:

a $6x^2 + 4x$ by first differentiating $x^3 + x^2$ **b** e^{3x+1} by first differentiating e^{3x+1}

c $\sqrt{x}$ by first differentiating $x\sqrt{x}$ **d** $(2x+1)^3$ by first differentiating $(2x+1)^4$

Definition

If $f(x)$ is a continuous positive function on an interval $a \leqslant x \leqslant b$ then the area under the curve between $x = a$ and $x = b$ is $\int_a^b f(x)\,dx$.



Definition

$$\int_a^t f(x)\,dx = F(t) - F(a) \quad \text{where} \quad F'(x) = f(x). \quad F(x) \text{ is the antiderivative of } f(x).$$

b) Fundamental theorem of calculus

Property

For a continuous function $f(x)$ with antiderivative $F(x)$,

$$\int_a^b f(x)\,dx = F(b) - F(a).$$

Property

- $\int_a^a f(x)\,dx = 0$
- $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$
- $\int_a^b f(x)\,dx + \int_b^c f(x)\,dx = \int_a^c f(x)\,dx$
- $\int_a^b [f(x) \pm g(x)]\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$
- $\int_a^b c\,dx = c(b-a)$ {c is a constant}
- $\int_a^b c\,f(x)\,dx = c\int_a^b f(x)\,dx$

Proof

$$\begin{aligned}\int_a^b f(x)\,dx + \int_b^c f(x)\,dx &= F(b) - F(a) + F(c) - F(b)\\ &= F(c) - F(a)\\ &= \int_a^c f(x)\,dx\end{aligned}$$

Example

Use the fundamental theorem of calculus to find the area between :

1. the x-axis and $y = x^2$ from $x = 0$ to $x = 1$
2. the x-axis and $y = \sqrt{x}$ from $x = 1$ to $x = 9$.

EXERCISE 9

Use the fundamental theorem of calculus to find the area between:

a the x-axis and $y = x^3$ from $x = 0$ to $x = 1$

b the x-axis and $y = x^2$ from $x = 1$ to $x = 2$

c the x-axis and $y = \sqrt{x}$ from $x = 0$ to $x = 1$.

Check your answers using technology.

EXERCISE 10

Use the fundamental theorem of calculus to show that:

a $\int_a^a f(x)\,dx = 0$ and explain the result graphically

b $\int_a^b c\,dx = c(b-a)$ where c is a constant

c $\int_b^a f(x)\,dx = -\int_a^b f(x)\,dx$

d $\int_a^b cf(x)\,dx = c\int_a^b f(x)\,dx$ where c is a constant

e $\int_a^b [f(x)+g(x)]\,dx = \int_a^b f(x)\,dx + \int_a^b g(x)\,dx$

EXERCISE 11

Use the fundamental theorem of calculus to find the area between the x-axis and:

a $y = x^3$ from $x = 1$ to $x = 2$

b $y = x^2 + 3x + 2$ from $x = 1$ to $x = 3$

c $y = \sqrt{x}$ from $x = 1$ to $x = 2$

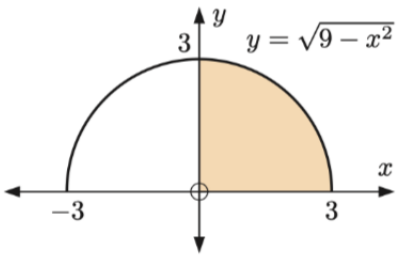
d $y = e^x$ from $x = 0$ to $x = 1.5$

e $y = \dfrac{1}{\sqrt{x}}$ from $x = 1$ to $x = 4$

f $y = x^3 + 2x^2 + 7x + 4$ from $x = 1$ to $x = 1.25$

EXERCISE 12

Use technology to find the area between the x-axis and $y = \sqrt{9-x^2}$ from $x = 0$ to $x = 3$.
Check your answer by direct calculation of the area.



III- Rules of integration

a) Indefinite integrals

Definition

This is process known as indefinite integration because it is not being applied to a particular interval.

If $F'(x) = f(x)$ then $\int f(x)\,dx = F(x) + c$.

Property

Any constant may be written in front of the integral sign.

$$\int k\,f(x)\,dx = k\int f(x)\,dx, \quad k \text{ is a constant}$$

Proof

Consider differentiating $kF(x)$ where $F'(x) = f(x)$.

$$\frac{d}{dx}\big(k\,F(x)\big) = k\,F'(x) = k\,f(x)$$

$$\therefore \quad \int k\,f(x)\,dx = k\,F(x)$$

$$= k\int f(x)\,dx$$

Property

The integral of a sum is the sum of the separate integrals. This rule enables us to integrate term by term.

$$\int [f(x) + g(x)]\,dx = \int f(x)\,dx + \int g(x)\,dx$$

Example

If $y = x^4 + 2x^3$, find $\frac{dy}{dx}$. Hence find $\int (2x^3 + 3x^2)\,dx$.

EXERCISE 13

If $y = x^3 + x^2$, find $\frac{dy}{dx}$. Hence find $\int (3x^2 + 2x)\,dx$.

EXERCISE 14

If $y = (2x+1)^4$ find $\frac{dy}{dx}$. Hence find $\int (2x+1)^3\,dx$.

Example

Suppose $y = \sqrt{5x-1}$.

1. Find $\frac{dy}{dx}$.
2. Hence find $\int \frac{1}{\sqrt{5x-1}}\,dx$.

Note. *We can check that an integral is correct by differentiating the answer. It should give us the integrand, the function we originally integrated.*

EXERCISE 15

If $y = \frac{1}{\sqrt{x}}$, find $\frac{dy}{dx}$. Hence find $\int \frac{1}{x\sqrt{x}}\,dx$.

EXERCISE 16

By considering $\frac{d}{dx}(x^2-x)^3$, find $\int (2x-1)(x^2-x)^2\,dx$.

EXERCISE 17

Find $\frac{dy}{dx}$ if $y = \sqrt{1-4x}$. Hence find $\int \frac{1}{\sqrt{1-4x}}\,dx$.

b) Antiderivatives

Derivative table summary

Function	*Derivative*	*Name*
c, a constant	0	
$mx+c$, m and c are constants	m	
x^n	nx^{n-1}	power rule
$cu(x)$	$cu'(x)$	
$u(x)+v(x)$	$u'(x)+v'(x)$	addition rule
$u(x)v(x)$	$u'(x)v(x)+u(x)v'(x)$	product rule
$\dfrac{u(x)}{v(x)}$	$\dfrac{u'(x)v(x)-u(x)v'(x)}{[v(x)]^2}$	quotient rule
$y=f(u)$ where $u=u(x)$	$\dfrac{dy}{dx}=\dfrac{dy}{du}\dfrac{du}{dx}$	chain rule
e^x	e^x	
$e^{f(x)}$	$e^{f(x)}f'(x)$	
$\ln x$	$\dfrac{1}{x}$	
$\ln f(x)$	$\dfrac{f'(x)}{f(x)}$	
$[f(x)]^n$	$n[f(x)]^{n-1}f'(x)$	
$\sin x$	$\cos x$	
$\cos x$	$-\sin x$	
$\tan x$	$\dfrac{1}{\cos^2 x}$	

Property

$$\int k\,dx = kx + c$$

Proof

For k a constant, $\dfrac{d}{dx}(kx+c)=k$

Property

$$\int x^n\,dx = \frac{x^{n+1}}{n+1}+c, \quad n\neq -1$$

Proof

If $n\neq -1$, $\dfrac{d}{dx}\left(\dfrac{x^{n+1}}{n+1}+c\right)=\dfrac{(n+1)x^n}{n+1}=x^n$

Property

$$\int e^x \, dx = e^x + c$$

Proof

$$\frac{d}{dx}(e^x + c) = e^x$$

Property

$$\int \frac{1}{x} \, dx = \ln|x| + c$$

Proof

If $x > 0$, $\quad \frac{d}{dx}(\ln x + c) = \frac{1}{x}$

If $x < 0$, $\quad \frac{d}{dx}(\ln(-x) + c) = \frac{-1}{-x} = \frac{1}{x}$

Property

$$\int \cos x \, dx = \sin x + c$$

Proof

$$\frac{d}{dx}(\sin x + c) = \cos x$$

Property

$$\int \sin x \, dx = -\cos x + c$$

Proof

$$\frac{d}{dx}(-\cos x + c) = \sin x$$

Note. *c is an arbitrary constant called the constant of integration or integrating constant.*

Example

Find :

1. $\int (x^3 - 2x^2 + 5) \, dx$
2. $\int \left(\frac{1}{x^3} - \sqrt{x}\right) dx$

Example

Integrate with respect to x :

1. $2\sin x - \cos x$
2. $-\dfrac{2}{x} + 3e^x$

EXERCISE 18

Integrate with respect to x:

a $3\sin x - 2$ b $4x - 2\cos x$ c $\sin x - 2\cos x + e^x$

d $x^2\sqrt{x} - 10\sin x$ e $\dfrac{x(x-1)}{3} + \cos x$ f $-\sin x + 2\sqrt{x}$

EXERCISE 19

Find:

a $\int \left(\sqrt{x} + \frac{1}{2}\cos x\right)\, dx$ b $\int \left(2e^t - 4\sin t\right)\, dt$ c $\int \left(3\cos t - \frac{1}{t}\right)\, dt$

EXERCISE 20

Find y if:

a $\dfrac{dy}{dx} = 6$ b $\dfrac{dy}{dx} = 4x^2$ c $\dfrac{dy}{dx} = 5\sqrt{x} - x^2$

d $\dfrac{dy}{dx} = \dfrac{1}{x^2}$ e $\dfrac{dy}{dx} = 2e^x - 5$ f $\dfrac{dy}{dx} = 4x^3 + 3x^2$

c) Particular value given

Example

Find $f(x)$ given that $f'(x) = x^3 - 2x^2 + 3$ and $f(0) = 2$.

Example

Find $f(x)$ given that $f'(x) = 2\sin x - \sqrt{x}$ and $f(0) = 4$.

Example

Find $f(x)$ given that $f''(x) = 12x^2 - 4$, $f'(0) = -1$, and $f(1) = 4$.

EXERCISE 21

Find $f(x)$ given that:

a $f'(x) = 2x - 1$ and $f(0) = 3$

b $f'(x) = 3x^2 + 2x$ and $f(2) = 5$

c $f'(x) = e^x + \frac{1}{\sqrt{x}}$ and $f(1) = 1$

d $f'(x) = x - \frac{2}{\sqrt{x}}$ and $f(1) = 2$

EXERCISE 22

Find $f(x)$ given that:

a $f''(x) = 2x + 1$, $f'(1) = 3$, and $f(2) = 7$

b $f''(x) = 15\sqrt{x} + \frac{3}{\sqrt{x}}$, $f'(1) = 12$, and $f(0) = 5$

c $f''(x) = \cos x$, $f'(\frac{\pi}{2}) = 0$, and $f(0) = 3$

d $f''(x) = 2x$ and the points $(1, 0)$ and $(0, 5)$ lie on the curve.

d) Integrating $f(ax+b)$

Property

For a, b constants with $a \neq 0$, we have :

Function	*Integral*
e^{ax+b}	$\frac{1}{a}e^{ax+b}+c$
$(ax+b)^n,\ n \neq -1$	$\frac{1}{a}\frac{(ax+b)^{n+1}}{n+1}+c$
$\frac{1}{ax+b}$	$\frac{1}{a}\ln\lvert ax+b\rvert+c$
$\cos(ax+b)$	$\frac{1}{a}\sin(ax+b)+c$
$\sin(ax+b)$	$-\frac{1}{a}\cos(ax+b)+c$

Proof

Proof by differentiating.

Example

Find :

1. $\int (2x+3)^4\,dx$
2. $\int \frac{1}{\sqrt{1-2x}}\,dx$

Example

Find :

1. $\int \left(2e^{2x} - e^{-3x}\right) dx$
2. $\int \frac{4}{1-2x}\, dx$

Example

Integrate with respect to x :

$$2\sin(3x) + \cos(4x + \pi)$$

Example

Integrate $(2 - \sin x)^2$ with respect to x, using the identity $\sin^2 x = \frac{1}{2} - \frac{1}{2}\cos(2x)$.

EXERCISE 23

Find:

a $\int (2x+5)^3\,dx$ b $\int \frac{1}{(3-2x)^2}\,dx$ c $\int \frac{4}{(2x-1)^4}\,dx$

d $\int (4x-3)^7\,dx$ e $\int \sqrt{3x-4}\,dx$ f $\int \frac{10}{\sqrt{1-5x}}\,dx$

EXERCISE 24

Integrate with respect to x:

a $\sin(3x)$ b $2\cos(-4x)+1$ c $3\cos\left(\frac{x}{2}\right)$

d $3\sin(2x)-e^{-x}$ e $2\sin\left(2x+\frac{\pi}{6}\right)$ f $-3\cos\left(\frac{\pi}{4}-x\right)$

EXERCISE 25

Using the identities $\cos^2 x = \frac{1}{2}+\frac{1}{2}\cos(2x)$ and $\sin^2 x = \frac{1}{2}-\frac{1}{2}\cos(2x)$ to help you, integrate with respect to x:

a $\cos^2 x$ b $\sin^2 x$ c $1+\cos^2(2x)$

d $3-\sin^2(3x)$ e $\frac{1}{2}\cos^2(4x)$ f $(1+\cos x)^2$

EXERCISE 26

Find:

a $\int \left(2e^x+5e^{2x}\right)\,dx$ b $\int \left(3e^{5x-2}\right)\,dx$ c $\int \left(e^{7-3x}\right)\,dx$

d $\int \frac{1}{2x-1}\,dx$ e $\int \frac{5}{1-3x}\,dx$ f $\int \left(e^{-x}-\frac{4}{2x+1}\right)\,dx$

EXERCISE 27

Consider a function g such that $g''(x) = -\sin 2x$.
Show that the gradients of the tangents to $y = g(x)$ when $x = \pi$ and $x = -\pi$ are equal.

e) Substitution

Theorem

We can integrate functions of this form using the theorem

$$\boxed{\int f(u)\,\frac{du}{dx}\,dx = \int f(u)\,du}$$

Proof

Suppose $F(u)$ is the antiderivative of $f(u)$, so $\dfrac{dF}{du} = f(u)$

$$\therefore \quad \int f(u)\,du = F(u) + c \quad \ldots\ (1)$$

But $$\frac{dF}{dx} = \frac{dF}{du}\,\frac{du}{dx} \qquad \{\text{chain rule}\}$$
$$= f(u)\,\frac{du}{dx}$$

$$\therefore \quad \int f(u)\,\frac{du}{dx}\,dx = F(u) + c = \int f(u)\,du \quad \{\text{from (1)}\}$$

Example

$$\int (x^2+3x)^4(2x+3)\,dx$$
$$= \int u^4\,\frac{du}{dx}\,dx \qquad \left\{u = x^2+3x, \quad \frac{du}{dx} = 2x+3\right\}$$
$$= \int u^4\,du \qquad \left\{\text{replacing } \frac{du}{dx}\,dx \text{ by } du\right\}$$
$$= \frac{u^5}{5} + c$$
$$= \tfrac{1}{5}(x^2+3x)^5 + c$$

Example

Use substitution to find :

$$\int \sqrt{x^3+2x}\,(3x^2+2)\,dx$$

Example

Use substitution to find :

1. $\int \frac{3x^2+2}{x^3+2x}\,dx$
2. $\int xe^{1-x^2}\,dx$

Example

Integrate with respect to x :

1. $\cos^3 x \sin x$
2. $\frac{\cos x}{\sin x}$

EXERCISE 28

Integrate by substitution:

a $4x^3(2+x^4)^3$ b $\frac{2x}{\sqrt{x^2+3}}$ c $\frac{x}{(1-x^2)^5}$ d $\sqrt{x^3+x}\,(3x^2+1)$

EXERCISE 29

Find:

a $\int \frac{2x}{x^2+1}\,dx$ b $\int \frac{x}{2-x^2}\,dx$ c $\int \frac{2x-3}{x^2-3x}\,dx$ d $\int \frac{6x^2-2}{x^3-x}\,dx$

EXERCISE 30

Integrate by substitution:

a $\frac{\sin x}{\sqrt{\cos x}}$ b $\tan x$ c $\sqrt{\sin x}\cos x$ d $x\sin(x^2)$

IV- Definite integrals

Note. *It is common to write* $F(b) - F(a)$ *as* $\left[F(x)\right]_a^b$*, and so*

$$\boxed{\int_a^b f(x)\,dx = \left[F(x)\right]_a^b = F(b) - F(a)}$$

Example

Find $\int_1^3 (x^2 + 2)\,dx$.

Example

Evaluate :

1. $\int_0^{\frac{\pi}{3}} \sin x\,dx$
2. $\int_1^4 \left(2x + \frac{3}{x}\right) dx$

EXERCISE 31

Evaluate the following and check with your graphics calculator:

a $\int_0^1 x^3\,dx$ b $\int_0^2 (x^2 - x)\,dx$ c $\int_0^1 e^x\,dx$

d $\int_0^{\frac{\pi}{6}} \cos x\,dx$ e $\int_1^4 \left(x - \frac{3}{\sqrt{x}}\right)\,dx$ f $\int_4^9 \frac{x-3}{\sqrt{x}}\,dx$

g $\int_1^3 \frac{1}{x}\,dx$ h $\int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \sin x\,dx$ i $\int_1^2 (e^{-x}+1)^2\,dx$

j $\int_2^6 \frac{1}{\sqrt{2x-3}}\,dx$ k $\int_0^1 e^{1-x}\,dx$ l $\int_0^{\frac{\pi}{6}} \sin(3x)\,dx$

EXERCISE 32

Use the identity $\cos^2 x = \frac{1}{2} + \frac{1}{2}\cos(2x)$ to help evaluate $\int_0^{\frac{\pi}{4}} \cos^2 x\,dx$.

EXERCISE 33

Evaluate using technology:

a $\int_1^3 \ln x\,dx$ b $\int_{-1}^1 e^{-x^2}\,dx$ c $\int_{\frac{\pi}{4}}^{\frac{\pi}{6}} \sin(\sqrt{x})\,dx$

EXERCISE 34

Show that $\frac{4x+1}{x-1}$ may be written in the form $4 + \frac{5}{x-1}$.

Hence show that $\int_3^5 \frac{4x+1}{x-1}\,dx = 8 + 5\ln 2$.

EXERCISE 35

Write as a single integral:

a $\int_2^4 f(x)\,dx + \int_4^7 f(x)\,dx$ b $\int_1^3 g(x)\,dx + \int_3^8 g(x)\,dx + \int_8^9 g(x)\,dx$

EXERCISE 36

Given that $\int_{-1}^1 f(x)\,dx = -4$, determine the value of:

a $\int_1^{-1} f(x)\,dx$ b $\int_{-1}^1 (2 + f(x))\,dx$ c $\int_{-1}^1 2f(x)\,dx$

d k such that $\int_{-1}^1 kf(x)\,dx = 7$